Q9: Logic

- Due 12 Jan at 12:30
- Points 6
- Questions 6
- Available after 5 Dec 2025 at 12:00
- Time limit 60 Minutes
- Allowed attempts 5

This quiz is no longer available as the course has been concluded.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	8 minutes	4 out of 6

ⓘ Correct answers are hidden.

Score for this attempt: 4 out of 6
Submitted 7 Dec 2025 at 17:17
This attempt took 8 minutes.

Question 1
1 / 1 pts
Let α, β, γ be sentences in an arbitrary logic. Which of the following statements are true?

- [x] If $\alpha \models \beta$ and $\beta \models \alpha$ then $M(\alpha) = M(\beta)$.
- [x] If $\alpha \models \beta$ and $\beta \models \gamma$ then $\alpha \models \gamma$.
- [ ] If $\alpha \models \beta$ and $\beta \models \gamma$ then $\gamma \models \alpha$.
- [x] If α entails β and β entails γ and γ entails α then $\alpha \equiv \gamma$.

Question 2
1 / 1 pts
Which of the following propositional logic formulas are valid (i.e. a tautology)?

- [ ] $(A \Rightarrow B) \Rightarrow (A \vee B)$.
- [ ] $(A \Rightarrow B) \Rightarrow (\neg A \Rightarrow \neg B)$.
- [ ] $(A \Rightarrow B) \Rightarrow (B \Rightarrow A)$.

- [x] $(A \Rightarrow B) \Rightarrow (\neg B \Rightarrow \neg A)$.

Question 3

1 / 1 pts

Consider the following knowledge base with sentences S_1, S_2, S_3:

- S_1 : $Rain \vee Cloud$.
- S_2 : $Sun \vee Cloud$.
- S_3 : $\neg Rain \vee \neg Rainbow$.

Which is a correct way of using the resolution rule?

- (•) We take S_1 and S_3 and get $Cloud \vee \neg Rainbow$.
- ( ) We take S_2 and S_3 and get $\neg Rain \vee \neg Rainbow$.
- ( ) We take S_1 and S_2 and get $Rain \vee Sun$.
- ( ) We take S_1 and S_3 and get $Cloud$.

Incorrect

Question 4

0 / 1 pts

Consider a sentence $\forall x, y. (Foo(x) = Foo(y)) \Rightarrow (Bar(x, y) = Bar(x, x))$ in first order logic. What is true?

- ( ) Foo is a predicate, Bar is a function
- ( ) Foo is a function, Bar is a predicate
- (•) Foo is a predicate, Bar is a predicate
- ( ) Foo is a function, Bar is a function

Question 5

1 / 1 pts

Which of the following sentences are valid (a tautology, necessarily true)?

- [ ] $\forall x. \exists y. \neg(x = y)$.
- [x] $\forall x. \forall y. (x = y \vee \neg(x = y))$.
- [ ] $\exists x. \forall y. (x = y)$.
- [x] $\exists x. \exists y. (x = y)$.

Incorrect

Question 6

0 / 1 pts

Consider the following knowledge base KB that represents an attempt to formalize a republic. MP stands for a member of the parliament, GOV stands for a member of the government, PM stands for a prime minister:

- $\exists x.\ \mathrm{GOV}(x)$.
- $\exists x.\ \mathrm{PM}(x)$.
- $\forall x.\ \mathrm{PM}(x) \Rightarrow \mathrm{GOV}(x)$.
- $\forall x, y.\ \mathrm{PM}(x) \wedge \mathrm{PM}(y) \Rightarrow x = y$.
- $\forall x.\ \mathrm{GOV}(x) \Rightarrow (\mathrm{MP}(x) \vee \mathrm{PM}(x))$.

Which of the following sentences should be added in order for "The parliament has at least one member." to be become true in *every* model of KB?

No sentence needs to be added. "The parliament has at least one member" is already true in every model of KB.

$\forall x.\ MP(x) \Rightarrow GOV(x).$

$\forall x.\ MP(x) \Rightarrow GOV(x) \vee PM(x).$

$\exists x, y.\ x \neq y \wedge GOV(x) \wedge GOV(y).$

Quiz score: 4 out of 6